Naser Kazemi

Research Focus: Reinforcement Learning & Robotics, World Models, Generative Video

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Education

École polytechnique fédérale de Lausanne (EPFL) *Lausanne, Switzerland*
Master of Science in Computer Science, AI and Data Science Specialization Sep. 2025 – Present
GPA: 5.85/6, Transcript.

Sharif University of Technology *Tehran, Iran*
Bachelor of Science in Computer Engineering, Minor in Applied Mathematical Science Sep. 2020 – Jul. 2025
GPA: 19.17/20 (Major: 19.75/20), Transcript. Coursework: RL, Convex Optimization, High-Dimensional Probability, Information Theory.

Research Experience

EPFL, Learning Algorithms and Systems Laboratory (LASA) *Lausanne, Switzerland*
Research Assistant, Cross-Embodiment Constraint Learning (with Prof. Aude Billard) Sep. 2026 – Present
- Formulating a cross-embodiment constraint learning method that disentangles environmental obstacles from robot-specific kinematic/dynamic limits, enabling zero-shot constraint transfer across robots and environments.

Mila, Quebec AI Institute, Robotics and Embodied AI Lab (REAL) *Montréal, Canada*
Research Intern, Causal World Models for RL (with Prof. Glen Berseth) Feb. 2026 – Present
- Developing causal world models with event-predictive latent spaces that separate agent interventions from exogenous environment dynamics, improving counterfactual reasoning and credit assignment in procedurally generated RL tasks.

INSAIT, Institute for Computer Science, Artificial Intelligence and Technology *Sofia, Bulgaria*
Research Intern, World Models (with Prof. Luc Van Gool & Prof. Danda Pani Paudel) Mar. 2024 – Aug. 2025
- Built and open-sourced **GenieRedux** (GitHub), a PyTorch reimplementation of DeepMind's *Genie* world model, and extended it with trained RL agent exploration for trajectory collection (*NeurIPS 2024 D3S3 Workshop*, first author).
- Co-designed an uncertainty-driven exploration framework for training multi-environment world models, improving video fidelity and action controllability without demonstrations or environment rewards (*CVPR 2025*).
- Designed **StateSpaceDiffuser**, integrating Mamba state-space memory into diffusion world models; improved long-horizon PSNR by 25% over diffusion baselines in 2D/3D environments and introduced a temporal-consistency benchmark (*NeurIPS 2025*).

Max Planck Institute for Informatics (MPI-INF) *Saarbrücken, Germany*
Research Intern – Multi-Objective Evolutionary Optimization (with Prof. Vahid Babaei) Jul. 2023 – Sep. 2023
- Improved Pareto-front diversity across NSGA-II and NSGA-III by designing a Monte Carlo objective-space segmentation and custom non-dominated sorting algorithm, yielding substantial gains on high-dimensional benchmarks.

Publications

- StateSpaceDiffuser: Bringing Long-Context Content to Diffusion World Models. N. Savov, **N. Kazemi**, D. Zhang, D. P. Paudel, X. Wang, L. Van Gool. *NeurIPS 2025*. arXiv
- Exploration-Driven Generative Interactive Environments. N. Savov, **N. Kazemi**, M. Mahdi, D. P. Paudel, X. Wang, L. Van Gool. *CVPR 2025*. arXiv, Webpage
- Learning Generative Interactive Environments by Trained Agent Exploration. **N. Kazemi**, N. Savov, D. P. Paudel, L. Van Gool. *NeurIPS 2024 Workshop (D3S3)*. arXiv, Webpage

Selected Projects

- **GenieRedux** (PyTorch) – Open-source DeepMind *Genie* world model (ST-Transformer tokenizer, LAM & MaskGIT). GitHub
- **End-to-End Discrete Concept Discovery** (PyTorch) – Vis2Sym bridging visual perception and symbolic reasoning via VQ. GitHub
- **Massive Data Algorithms** (PySpark) – Distributed streaming, graph analysis, and high-dimensional clustering on Spark. GitHub
- **C-Minus Compiler** (Python) – Lexer, parser, semantic analyzer, and code generator for a C subset. GitHub

Technical Skills

Programming	Python, C/C++, CUDA, Java, Scala, R
ML & Data	PyTorch, JAX, TensorFlow, scikit-learn, Spark, CVXPY
Robotics & Tools	Franka Emika, KUKA IIWA, ROS 2, Isaac Sim, MuJoCo, Linux, Git, Docker, Slurm, WandB
Languages	English (IELTS 8.0), Persian (Native)